

Oblique Collisions Between Two Spheres

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1 Introduction

Consider two spheres of masses m_1 and m_2 , with initial velocities $\mathbf{u}_1$ and $\mathbf{u}_2$. During collision, the impulse acts along the *line of centres*, which we denote by the unit vector $\mathbf{I}$. Only the components of velocity parallel to $\mathbf{I}$ change during the collision; perpendicular components remain unchanged.

2 Velocity Decomposition

For each sphere, decompose the velocity into components parallel and perpendicular to the impulse direction:

$$\mathbf{u}_i = u_{i\parallel} \mathbf{I} + \mathbf{u}_{i\perp}, \quad i = 1, 2.$$

After the collision,

$$\mathbf{v}_i = v_{i\parallel} \mathbf{I} + \mathbf{u}_{i\perp}.$$

3 Newton's Law of Restitution

Newton's law of restitution applies only to the parallel components:

$$e = \frac{v_{2\parallel} - v_{1\parallel}}{u_{1\parallel} - u_{2\parallel}},$$

where e is the coefficient of restitution.

Rearranging:

$$v_{2\parallel} - v_{1\parallel} = -e(u_{2\parallel} - u_{1\parallel}).$$

4 Conservation of Momentum

Linear momentum parallel to $\mathbf{I}$ is conserved:

$$m_1 u_{1\parallel} + m_2 u_{2\parallel} = m_1 v_{1\parallel} + m_2 v_{2\parallel}.$$

5 Solving for the Final Velocities

Solving the system consisting of restitution and momentum conservation gives the standard 1D collision formulas:

$$v_{1\parallel} = \frac{m_1 u_{1\parallel} + m_2 u_{2\parallel} - m_2 e(u_{1\parallel} - u_{2\parallel})}{m_1 + m_2},$$
$$v_{2\parallel} = \frac{m_1 u_{1\parallel} + m_2 u_{2\parallel} + m_1 e(u_{1\parallel} - u_{2\parallel})}{m_1 + m_2}.$$

6 Reconstructing the Final Velocity Vectors

The final velocity vectors are:

$$\mathbf{v}_1 = v_{1\parallel} \mathbf{I} + \mathbf{u}_{1\perp}, \quad \mathbf{v}_2 = v_{2\parallel} \mathbf{I} + \mathbf{u}_{2\perp}.$$

7 Computing the Impulse Direction

Given sphere centres at (x_1, y_1) and (x_2, y_2) , the line of centres is:

$$\mathbf{I} = \frac{(x_2 - x_1, y_2 - y_1)}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}.$$

The parallel components are:

$$u_{i\parallel} = \mathbf{u}_i \cdot \mathbf{I}, \quad \mathbf{u}_{i\perp} = \mathbf{u}_i - u_{i\parallel} \mathbf{I}.$$

8 Collision Condition

A collision occurs when the distance between centres satisfies:

$$d = 2R,$$

where R is the radius of each sphere.

9 Conclusion

This formulation allows efficient computation of post-collision velocities for oblique collisions in a 2D physics engine. Only the parallel components change, and the perpendicular components remain unaffected.